

3D Scatter Plot Projection Project Report

1. Project Overview

This project visualizes randomly generated 3D points on a 2D canvas using orthographic and perspective projection. Users can rotate the scene using yaw, pitch and roll.

2. Workflow

Random 3D Points → Store in JSON → Load in JavaScript → Convert Degrees to Radians → Apply Pitch, Yaw and Roll → Translate Along Z → Apply Projection → Draw on Canvas → Animate

Random	JSON	Rotate	Project	Canvas
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3. Random Point Generation

Each point is generated with random x, y and z values and stored in data.json.

4. Degree to Radian

Radians = Degrees $\times \pi / 180$. Trigonometric functions require radians.

5. Rotation Equations

Pitch: $y' = y \cos \theta - z \sin \theta$, $z' = y \sin \theta + z \cos \theta$

Yaw: $x' = x \cos \theta + z \sin \theta$, $z' = -x \sin \theta + z \cos \theta$

Roll: $x' = x \cos \theta - y \sin \theta$, $y' = x \sin \theta + y \cos \theta$

6. Orthographic Projection

$X=x$, $Y=y$. Depth is ignored.

7. Perspective Projection

$Scale = f / (f + z)$, $X = x \times Scale$, $Y = y \times Scale$. Distant objects appear smaller.

8. Implementation

HTML provides controls, CSS styles the page, JavaScript loads data, rotates points, projects them and renders frames using `requestAnimationFrame()`.

9. Conclusion

The project demonstrates the complete 3D graphics pipeline from point generation to interactive visualization.